Dr. Madhurima Panja

Country of Birth: India
Citizenship: Indian
Residency: UAE Golden Visa | Location: Abu Dhabi

Lecturer & Data Scientist
AI | Data Science | Machine Learning
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Areas of Expertise

Artificial Intelligence, Machine Learning and Deep Learning, Data Science, Statistical Learning, Time Series and Spatiotemporal Modeling, Geometric Deep Learning, Generative AI, Python.

Academic & Professional Experience

Sep 2025 – Present **Data Scientist**, *Statistics Centre – Abu Dhabi, Department of Government Enablement*, Abu Dhabi, UAE.
- Forecasting and ML models for inflation, labour market, GDP, trade, and population.
- Data-engineering workflows for ingestion, validation, and model-ready datasets.
- Decision-support systems using scenario analysis and economic/administrative dashboards.

Jan 2023 – Present **Lecturer and Affiliated Researcher**, *Sorbonne University Abu Dhabi*, Abu Dhabi, UAE.
- **Teaching:** Undergraduate teaching in machine learning, statistics, and data science.
- **Executive education:** ADIA programs in AI, Python, statistics, and forecasting.
- **Research:** ML forecasting, geometric deep learning, and probabilistic modelling.

2022 **Graduate Teaching Assistant**, IIIT Bangalore, India.

2020 – 2021 **Research Associate**, Indian Statistical Institute, India.

Apr 2019 – Jul 2019 **Data Analyst Intern**, Alteus Biogenics Pvt. Ltd., India.

Education

2022–2025 Doctor of Philosophy (Ph.D.) in Data Science & AI, **IIIT Bangalore**, India.
- Thesis Title: On the design of neural networks for time series forecasting.

2018–2020 Master of Science in Statistics, **University of Madras**, India.
- *Gold Medalist* (Ranked 1st in the university).

2015–2018 Bachelor of Science in Statistics, **University of Calcutta**, India.
- Major in Statistics and Minor in Mathematics and Economics.

Teaching, Curriculum & Academic Coordination

1. Developed lecture materials, tutorials, assignments, and assessments for undergraduate quantitative and computing courses.
2. Designed Python-based practical exercises linking theoretical concepts with real-world datasets and applications.
3. Supervised and mentored students on coursework and applied data-analysis projects.
4. Designed and hosted Kaggle-based competitions across multiple courses to promote applied problem-solving and engagement with real-world datasets.

Teaching Experience

Sorbonne University Abu Dhabi

2023 – 2025 MATH370 - Data Science and Machine Learning

2023 – 2024 MATH350 - Statistical Inference

2023 – 2025 MATH260 - Multivariate Data Analysis

2024 CPSC001 - Computer Science

IIIT Bangalore, India

2022 Mathematics for Machine Learning

2022 Spatial Computing

Industry & Executive Education

Executive Program – Abu Dhabi Investment Authority (ADIA)

2024 – 2026 AI for Finance - Business Forecasting

2023 – 2026 AI for Finance - Descriptive Statistics with Python

2023 – 2025 AI for Finance - Python II

2023 – 2025 A Guided Tour of AI

Selected Publications (*Joint First Author, Citations: 480+, h-index: 8)

Submitted Papers

Full Publication list - Google Scholar.

[16] **Panja, M.**, D'Agostino, D., Li, H., Chakraborty, T., and Liu, N., EpiCastBench: Datasets and Benchmarks for Multivariate Epidemic Forecasting, Submitted to *NeurIPS*, 2026 (arXiv).

[15] **Panja, M.**, Younes, G., and Chakraborty, T., Turning Time Series into Algebraic Equations: Symbolic Machine Learning for Interpretable Modeling of Chaotic Time Series, Submitted, 2026 (arXiv).

[14] Besher, D., Pathak, R., **Panja, M.**, and Chakraborty, T., Beyond Point Forecasts: A Survey on Probabilistic Forecasting for Time Series and Spatiotemporal Data, Submitted to *TMLR*, 2026.

[13] Chakraborty, T., Besher, D., **Panja, M.**, and Sengupta, S., Neural ARFIMA model for forecasting BRIC exchange rates with long memory under oil shocks and policy uncertainties, Submitted, 2025 (arXiv).

Journal Publications

[12] **Panja, M.**, Chakraborty, T., Biswas, A., Deb, S., ESTGCN: Extreme Spatiotemporal Graph Convolutional Networks for Air Quality Forecasting, In Press *Journal of the Royal Statistical Society Series A*, 2026 (DOI).

[11] **Panja, M.**, Das, D., Ray, A., Chakraborty, T., Athulya, R., Hens, C., Dana, S. K., Murukesh, N., and Ghosh, D., Forecasting precipitation in the Arctic using probabilistic machine learning informed by causal climate drivers, *Chaos*, Vol. 35, article number 113108, 2025 (DOI).

[10] Barman, M.*, **Panja, M.***, Mishra, N., Chakraborty, T., Epidemic-guided deep learning for spatiotemporal forecasting of Tuberculosis outbreak, *Machine Learning*, Vol. 114, article number 213, 2025 (DOI).

[9] Naik, S.M., Chakraborty, T., **Panja, M.**, Hadid, A., Chakraborty, B., Skew Probabilistic Neural Networks for Learning from Imbalanced Data, *Pattern Recognition*, Vol. 165, pg. 1-18, 2025 (DOI).

[8] Chakraborty, T., Reddy U., Naik, S.M., **Panja, M.**, Manvitha, B., Ten Years of Generative Adversarial Nets (GANs): A survey of the state-of-the-art, *Machine Learning: Science and Technology*, Vol. 5, pg. 1-35, 2024 (DOI).

[7] Jakhmola, Y.*, **Panja, M.*** Mishra, N.K., Ghosh, K., Kumar, U., Chakraborty, T., Spatiotemporal Forecasting of Traffic Flow using Wavelet-based Temporal Attention, *IEEE Access*, Vol. 12, pg. 188797-188812, 2024 (DOI).

[6] **Panja, M.**, Chakraborty, T., Kumar, U., Liu, N., Epicasting: An Ensemble Wavelet Neural Network for Forecasting Epidemics, *Neural Networks*, Vol. 165, pg. 185-212, 2023 (DOI).

[5] **Panja, M.**,Chakraborty, T., Nadim. Sk., Ghosh. I., Kumar, U., Liu, N., An ensemble neural network approach to forecast Dengue outbreak based on climatic condition, *Chaos, Solitons & Fractals*, Vol. 167, pg. 1-14, 2023 (DOI).

Conference Publications

[4] **Panja, M.**, Modak, O., Younes, G., and Chakraborty, T., Zero-shot forecasting of epidemics, In *Advances in Neural Information Processing Systems (NeurIPS) - BERT2S*, 2025 (link).

[3] Thottolil, R., **Panja, M.**, Kumar, U., Analysis of Regional Air Quality Data using Extreme Value Theory and Deep Learning Models, *In IGARSS*, 2025 (link).

[2] Dutta, A*., **Panja, M*.**, Kumar, U., Hens, C., Chakraborty, T., Van der Pol-informed Neural Networks for Multi-step-ahead Forecasting of Extreme Climatic Events, *In Advances in Neural Information Processing Systems (NeurIPS) - AI4Science*, 2023 (link).

[1] **Panja, M.,** Chakraborty, T., Kumar, U., Hadid, A., Probabilistic AutoRegressive Neural Networks for Accurate Long-Range Forecasting, *In ICONIP*, 2023 (link).

Awards and Fellowships

2022 – 2024 Research Fellowship by IIIT Bangalore, India

2021 Thiru K.Kamaraj Gold Medal by University of Madras, India

2021 Prof. C.R.Rao and Bhargavi Rao Endowment Medal by University of Madras, India

2021 Prof. D.V.Raja Lakshman Endowment Medal by University of Madras, India

2018 – 2020 Merit Fellowship for M.Sc. Statistics by University of Madras, India

Research Software

R - *epicasting* - Ewnet: An Ensemble Wavelet Neural Network for Forecasting (Link)

R - *narfima* - Neural AutoRegressive Fractionally Integrated Moving Average Model (Link)

Python - *genformer* - Deep Generative Transformers (Link)

Technical Skills

Software R, Python, Matlab, Minitab, SPSS, LaTeX, Alteryx, SQL

Library TensorFlow, Pytorch, Darts, Scikit-learn

Computing Google Colab, Google Cloud Platform, Microsoft Azure

References

Prof. Dibakar Ghosh
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